



UNIVERSITAS
GADJAH MADA

The Implementation of C++ for Daily Habit Reminder and Management

c++ based app that helps the user implement
a new habit and turning it into daily drives

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Group member :

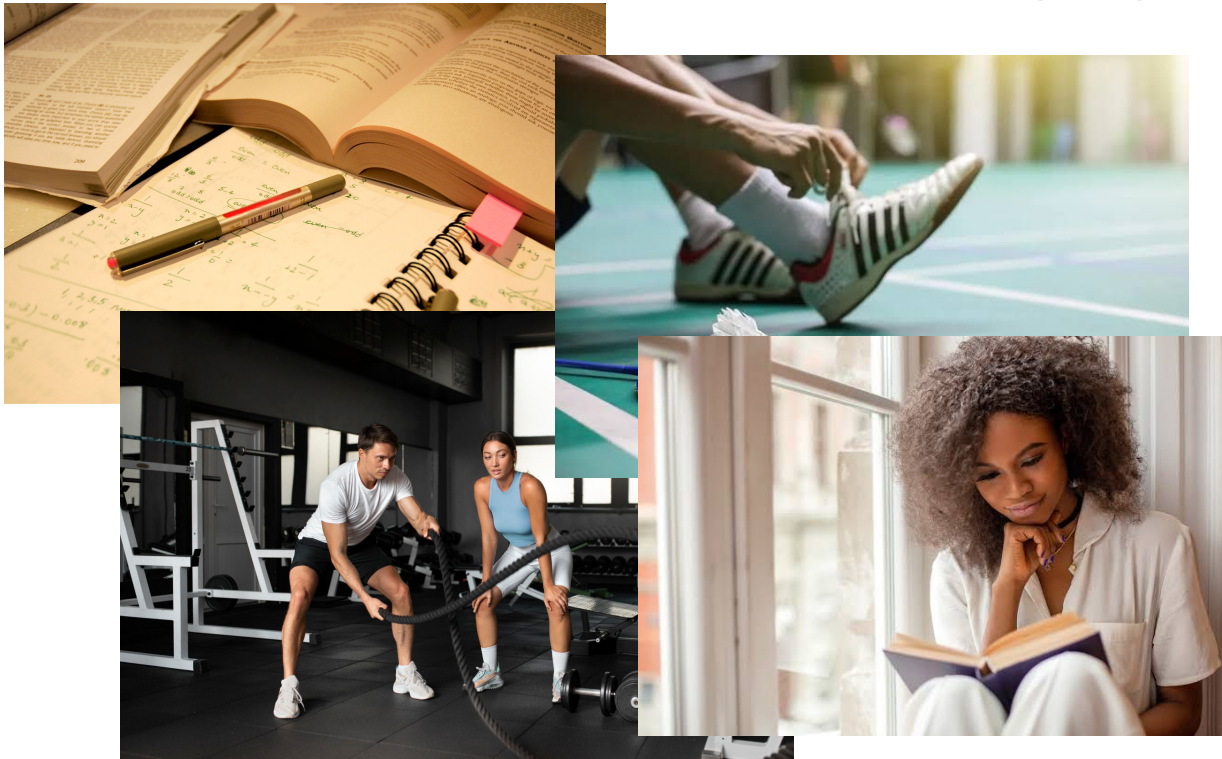
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How many good habit have you create this month ?



How many of them have u maintained throughout the month ?

2026



January	February	March
S M T W T F S	S M T W T F S	S M T W T F S
1 2 3	1 2 3 4 5 6 7	1 2 3 4 5 6 7
4 5 6 7 8 9 10	8 9 10 11 12 13 14	8 9 10 11 12 13 14
11 12 13 14 15 16 17	15 16 17 18 19 20 21	15 16 17 18 19 20 21
18 19 20 21 22 23 24	22 23 24 25 26 27 28	22 23 24 25 26 27 28
25 26 27 28 29 30 31		29 30 31
April	May	June
S M T W T F S	S M T W T F S	S M T W T F S
1 2 3 4	1 2	1 2 3 4 5 6
5 6 7 8 9 10 11	3 4 5 6 7 8 9	7 8 9 10 11 12 13
12 13 14 15 16 17 18	10 11 12 13 14 15 16	14 15 16 17 18 19 20
19 20 21 22 23 24 25	17 18 19 20 21 22 23	21 22 23 24 25 26 27
26 27 28 29 30	24 25 26 27 28 29 30	28 29 30
	31	
July	August	September
S M T W T F S	S M T W T F S	S M T W T F S
1 2 3 4	1	1 2 3 4 5
5 6 7 8 9 10 11	2 3 4 5 6 7 8	6 7 8 9 10 11 12
12 13 14 15 16 17 18	9 10 11 12 13 14 15	13 14 15 16 17 18 19
19 20 21 22 23 24 25	16 17 18 19 20 21 22	20 21 22 23 24 25 26
26 27 28 29 30 31	23 24 25 26 27 28 29	27 28 29 30
	30 31	
October	November	December
S M T W T F S	S M T W T F S	S M T W T F S
1 2 3	1 2 3 4 5 6 7	1 2 3 4 5
4 5 6 7 8 9 10	8 9 10 11 12 13 14	6 7 8 9 10 11 12
11 12 13 14 15 16 17	15 16 17 18 19 20 21	13 14 15 16 17 18 19
18 19 20 21 22 23 24	22 23 24 25 26 27 28	20 21 22 23 24 25 26
25 26 27 28 29 30 31	29 30	27 28 29 30 31

Printable Calendars by Betacalendars.com

What are the obstacle ?

Not Feeling
Excited ?

There is no
gratification ?

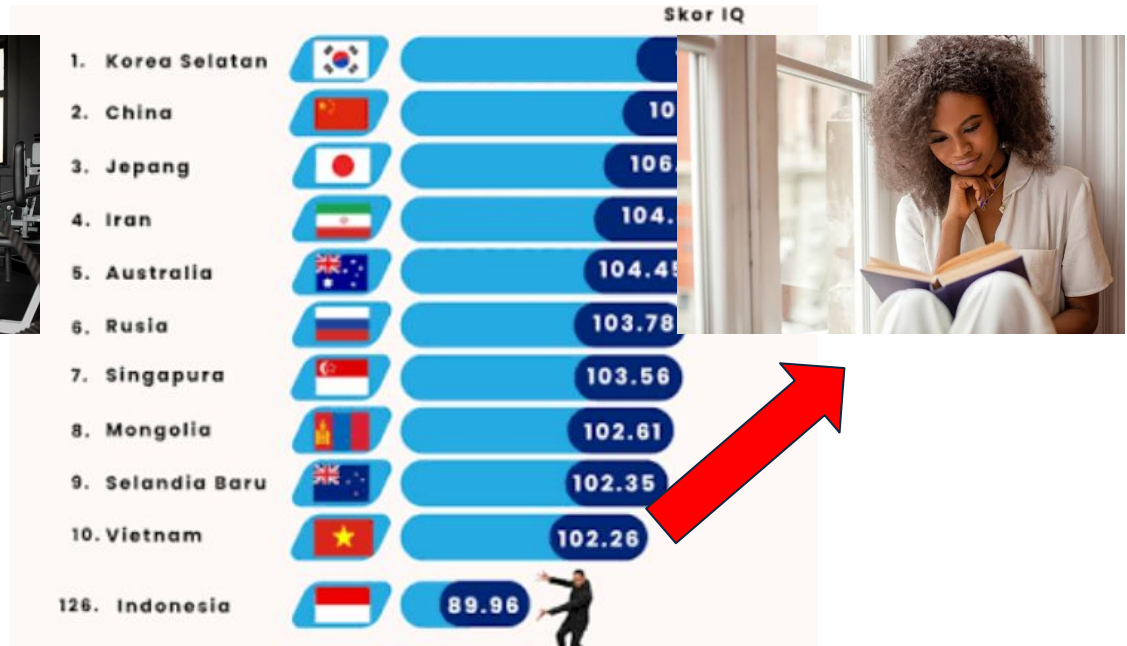
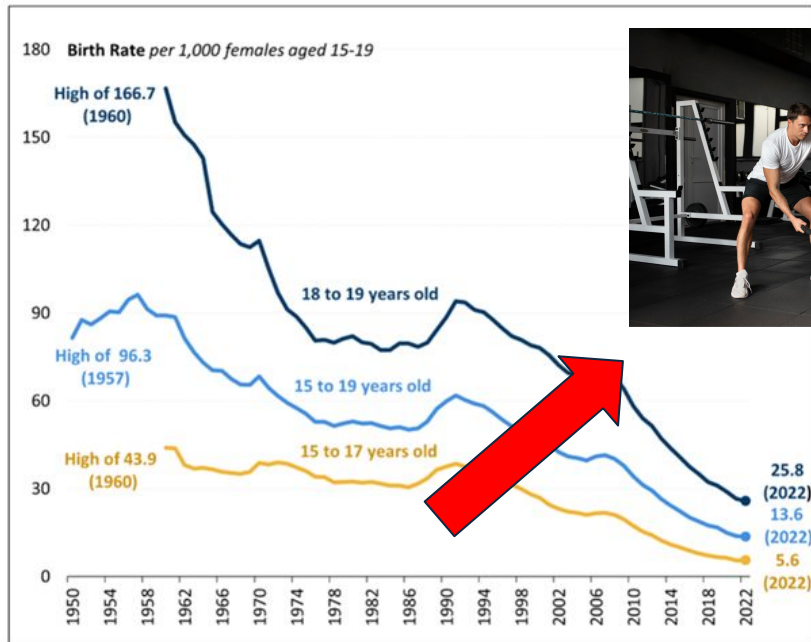
Feels very
lazy ?



The Reason
Doesn't Held up ?

No motivation to
do it ?

Not just about habits, but about our future



The data shows that teenagers' health is declining.

Our country's IQ score is still relatively low.

Our solution is : Habit tracker with maintaining capabilities



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How The App Works



The user input the habit



Assign the habit to a designated time (00.00 to 24.00) or linked into an existing habits



Track the existing habit via graph/streak

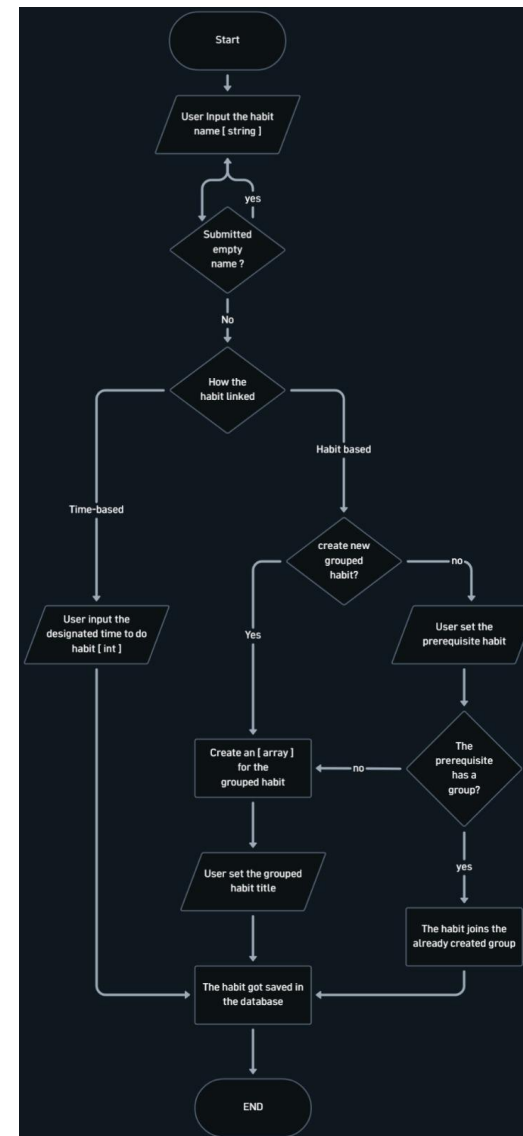


if the user passed the habits , the app will trigger the penalty system

The Flowchart

Start ↓ User Input the habit name [string] ↓ Submitted empty name?

- Yes → Go back to: User Input the habit name [string]
- No ↓ How the habit linked
- Option A: Time-based ↓
 - User input the designated time to do habit [int] ↓
 - Proceed to: The habit got saved in the database
- Option B: Habit based ↓
 - create new grouped habit?
 - Yes ↓
 - Create an [array] for the grouped habit ↓
 - User set the grouped habit title ↓
 - Proceed to: The habit got saved in the database
 - No ↓
 - User set the prerequisite habit ↓
 - The prerequisite has a group?
 - No → Go to: Create an [array] for the grouped habit
 - Yes ↓
 - The habit joins the already created group ↓
 - Proceed to: The habit got saved in the database ↓ The habit got saved in the database ↓ END



Key Features :

1. Compact
2. Habit Track Record
3. Activities (grouped habit)
4. App notification and reminder
5. Streak System

6.



PENALTY SYSTEM??

This app has a special feature: Penalty System. The main purpose of this system is to prevent the user to not forgetting about their own habits, so it makes the users could be more discipline, more consistent and not forgetting the things that they should do.

The penalty system will be in a form of **reflection and evaluation** mainly consists: Why did u forget to do your own habits? What preventing you to forget to do your own habits? What is your concrete act so you will never or decreased the amount of negligence for your own sake.

The user will write this reflection for a minimum of 100 words and the user can't do anything else if they haven't filled out the reflection.

Advantages

1. Easy to navigate system
2. Run locally
3. Offline
4. Direct visual cue

Disadvantages

1. Doesn't have cloud stored data
2. long development
3. Heavy UI effort
4. Non cross-platform

Source's :

1. Gardner, B. (2015). A review and analysis of the use of 'habit' in understanding, predicting and influencing health-related behaviour. *Health Psychology Review*, 9(3), 277–295. <https://doi.org/10.1080/17437199.2013.876238>
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3. Milkman, K. L., Minson, J. A., & Volpp, K. G. (2014). Holding the Hunger Games hostage at the gym: An evaluation of temptation bundling. *Management Science*, 60(2), 283–299. <https://doi.org/10.1287/mnsc.2013.1784>
4. Wood, W., & Rünger, D. (2016). Psychology of habit. *Annual Review of Psychology*, 67(1), 289–314. <https://doi.org/10.1146/annurev-psych-122414-033417>



Thank you for listening!

Any Questions?

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